Teaching to the Test: Detecting Test Gaming by AI Agents in Autonomous Scientific Discovery

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Problem and Motivation

AI research agents, programs that write and run code on their own, increasingly perform research, and their work is often judged by its tests. This creates a risk we call *test gaming*: an agent that can see the tests may make them pass without solving the problem, e.g., by returning expected answers directly [1]. Research code is especially exposed: its correct outputs are often unknown (the *test oracle problem*), making independent checks essential.

Goals and Research Questions

- **RQ1:** How often do agents game visible tests, and does hiding them help?
- **RQ2:** Which hidden checks best detect gamed code?
- **RQ3:** When reviewing its own code, does the agent admit to gaming?

Approach and Anticipated Results

We will select or adapt 6–10 tasks from DiscoveryBench [2] and ScienceAgentBench [3], requiring Python code to analyse scientific datasets and produce checkable results, such as statistics or predictions. A reusable harness will run the same coding agent repeatedly on each task with all, some, or none of the tests visible, using equal execution budgets. A separate evaluation suite will always remain hidden. We will compare hidden unit tests (pytest) and metamorphic tests using Hypothesis (e.g., row-order invariance for order-independent analyses); mutation testing (mutmut) will assess test-suite strength. Hidden-test failures flag suspected gaming, which code and log review will distinguish from ordinary faults. We will compare gaming rates, detection rates, and agent self-reviews against our classifications. We expect visible tests to encourage gaming, metamorphic checks to detect additional cases, and self-reviews to miss some cases.

References

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3. Chen, Z., et al.: ScienceAgentBench: Toward rigorous assessment of language agents for data-driven scientific discovery. In: Proc. ICLR (2025)